THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

1. Executive summary

Floods displace millions of people in India each year. The limiting factor in a rescue is locating survivors, not reaching them.

Three one-metre, 6.4 kg multirotor aircraft search flooded ground as a coordinated group under one operator, identify people from the air, report their position to a metre, and deliver a relief kit to each. No mobile network or internet is required.

Sizing, error budgets and flight software are complete; coordination is demonstrated across three simulated autopilots. No aircraft has flown, so the figures here are calculated.

We ask for INR 7,12,210, released in 30 steps against stated results rather than as one grant. Every part is a current listing from a named Indian supplier, so this is a priced basket and not an estimate. The first release is INR 3,605, and nothing above INR 26,000 is committed until a measured thrust figure exists.

2. The problem

Boat teams search slowly, at risk, and cannot see rooftops. Helicopters are scarce and committed to evacuation. Conventional drones need one pilot each, depend on infrastructure the flood has removed, and cannot assist a survivor once found.

Finding people is the harder problem. Someone in floodwater shows only head and shoulders, about 40 cm across. Detection software works on a small, fixed picture size, so the usual approach shrinks the photograph — and the person with it — until they are too small to recognise reliably. Choosing a smaller, cheaper model makes this worse rather than better.

There is also no public collection of labelled flood photographs to learn from, and thermal cameras do not help, because someone in water is close to the temperature of the water around them.

3. Our solution

System. The operator draws a search area of any shape. The software divides it into three fair shares and plans a route over each; we have tested this on thirteen awkward shapes, and no route strays outside the area. The aircraft then fly themselves — our tests confirm that no further command is sent once a mission has started. Each one recognises people and works out where they are without help from the ground. It carries four relief kits, comes down over a survivor, stops still, and drops one. A lost radio link, a low battery and the operator pressing abort all lead to the same thing: come home and land.

Detection. Rather than shrink the picture to fit the detector, we cut each frame into tiles and search them at full resolution, and we fly lower. Together those two changes make a person in the water **nine times larger** in the image, which lifts them out of the size range where these detectors stop working. Either change on its own is not enough. The cost is four times the computing, which the chosen accelerator has, and the mission is no longer than before because the lower flight saves more climbing than the extra passes cost.

Open question. Confirming a sighting across several frames would need half as much computing again as we have. There is a way to recover it, by skipping the tiles that are plainly open water, but a filter that throws away a tile containing a person fails silently. We will test it against public data before adopting it.

4. Deliverables

1. Three aircraft operating as a coordinated group under one operator.
2. A ground station that shows what is happening and, by design, cannot fly the aircraft.
3. The autonomy software, with an automatic test suite behind it.
4. Measured detection accuracy and a verdict on the two-stage filter.
5. A publication: tiling, aerial person detection and low-power onboard inference have not been combined for flood conditions.
6. Source code and results, published in either outcome.

5. Benefits

- **Operational.** Ten hectares in under eight minutes, returning coordinates rather than a video feed needing continuous attention.
- **Economic.** The complete system costs less than one hour of helicopter search time, travels by road, and is operated by two people.
- **Institutional.** A thrust stand, ground station, tested codebase and trained students remain with the department.
- **Strategic.** Indian suppliers wherever they meet the specification; domestic lead times are half those of imports.

6. Resources needed

Funding is requested in 30 small steps rather than as one grant. Each step buys one named set of parts and is released only once the previous step has produced a stated result. If a step fails, the programme stops there and nothing beyond it has been spent.

The order follows engineering dependency: one motor is measured before the other eleven are bought, and the first aircraft flies a complete mission before the second is begun.

Phase schedule

The third column is the amount released. The fourth is what must already be true before it is released.

#	Objective	INR	Released only after
1	Establish thrust and mass measurement capability	3,605	Approval
2	Confirm one motor lifts what the design needs	5,408	Instruments commissioned
3	Aircraft 1 — autopilot and control link	25,541	**Thrust verified**
4	Aircraft 1 — onboard computer and AI accelerator	41,694	Autopilot bench-tested
5	Aircraft 1 — centimetre positioning (RTK rover)	33,925	Compute stack operating
6	Aircraft 1 — camera and the three radios	16,363	Position fix acquired
7	Aircraft 1 — airframe fabrication	15,557	Avionics integrated
8	Aircraft 1 — battery pack and power distribution	35,527	Airframe fabricated
9	Aircraft 1 — speed controllers, release servos, wiring	9,042	Pack bench-discharged
10	Aircraft 1 — propulsion completed	23,774	Drive train installed
11	Charging, pack instrumentation and manual override	37,865	Aircraft assembled and weighed
12	Field consumables and assembly materials	7,934	Charging verified

#	Objective	INR	Released only after
13	Aircraft 2 — autopilot and control link	25,541	**Aircraft 1 flew a full mission**
14	Aircraft 2 — onboard computer and AI accelerator	41,694	Autopilot bench-tested
15	Aircraft 2 — centimetre positioning (RTK rover)	33,925	Compute stack operating
16	Aircraft 2 — camera and the three radios	16,363	Position fix acquired
17	Aircraft 2 — airframe fabrication	14,466	Avionics integrated
18	Aircraft 2 — battery pack and power distribution	26,028	Airframe fabricated
19	Aircraft 2 — speed controllers, release servos, wiring	8,871	Pack bench-discharged
20	Aircraft 2 — propulsion completed	29,182	Drive train installed
21	Aircraft 3 — autopilot and control link	25,541	**Two aircraft, separation verified**
22	Aircraft 3 — onboard computer and AI accelerator	41,694	Autopilot bench-tested
23	Aircraft 3 — centimetre positioning (RTK rover)	33,925	Compute stack operating
24	Aircraft 3 — camera and the three radios	16,363	Position fix acquired
25	Aircraft 3 — airframe fabrication	14,471	Avionics integrated
26	Aircraft 3 — battery pack and power distribution	26,028	Airframe fabricated
27	Aircraft 3 — speed controllers, release servos, wiring	8,871	Pack bench-discharged
28	Aircraft 3 — propulsion completed	36,731	Drive train installed
29	RTK base receiver, the source of the corrections	33,925	Third aircraft flying
30	Ground station: three video feeds, data link, base mount	22,351	Base receiver acquired
	Total	**7,12,210**	

Phase 3 commits the first significant expenditure and is released only on a measured thrust figure. Phase 13 begins the second aircraft only after the first has flown a complete mission, so the largest blocks of expenditure fund replication of a demonstrated configuration rather than a projection.

Component selection

The schedule gives the amount *released* per phase. The tables below give the parts each phase *buys*. A released amount is the parts cost plus tax where tax is still owed, plus 15 % contingency. Across the programme that is INR 5,95,232 of parts against INR 7,12,210 released.

Every line below is a current listing from a named Indian supplier, priced and linked. An earlier version of this brief said the opposite — that a handful of lines were firm and the rest were estimates. That is no longer true of any line, and the schedule and the tables are generated from the same list, so they cannot drift apart.

Three corrections are worth stating plainly, because they move the figure in both directions and together they *reduced* the ask by about INR 31,000. The onboard computer was budgeted at INR 8,000 against a real price of INR 19,999. The certified scale is no longer in the ask at all, because the department already owns one. And the earlier version added customs duty and GST to every line, which was right when the lines were estimates but wrong for Indian retail prices, where the tax is already paid.

Terms used below. *RTK*: satellite positioning corrected against a second receiver on a surveyed point, giving centimetre rather than metre accuracy. *6S3P*: eighteen cells, six in series for voltage and three in parallel for capacity. *kgf*: thrust written as the weight it would lift. *TOPS*: trillion operations per second, the accelerator's throughput. *ESC*: the electronic speed controller between battery and motor. *BMS*: the board that balances and protects the cells — there is none in this build, and the note after the tables says why.

Measurement instruments — phases 1–2 INR 3,135 of parts

Item	Model	INR, all 3	Why this part, and from whom
Load-cell amplifier	7Semi HX711 24-bit ADC	104	Turns the load cell's signal into a reading we can log. *Robu*

Item	Model	INR, all 3	Why this part, and from whom
Load cell	REES52 20 kg load cell + HX711 breakout	854	The motors ship without a thrust curve, so we measure it ourselves. Ready-made stands cost around ten times this. *Amazon India*
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	373	Holds the motor clear of the bench, so its own draught does not flatter the reading. *Flipkart*
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	1,140	A carbon airframe is bolted, not glued. *OnlyScrews*
Threadlocker	Loctite 243, 50 ml	664	Stops bolts working loose under motor vibration. Medium strength, so a joint can still be undone. *Amazon India*

Per aircraft, avionics and sensing — phases 3–6, repeated at 13–16 and 21–24 INR 81,399 per aircraft, INR 3,44,197 in all

Item	Model	INR, all 3	Why this part, and from whom
Flight controller	Holybro Pixhawk 6C Mini, 3 off	66,147	The autopilot: flies the aircraft and runs its safety behaviours. Previously a quotation, now ordinary retail. *Robu*
FC vibration mount	Glass-fibre anti-vibration shock absorber, 3 off	483	Isolates the autopilot from propeller vibration, which otherwise confuses its sensors. *Robu*
Companion computer	Raspberry Pi 5, 8 GB, 3 off	59,997	The computer the autonomy runs on. The accelerator and the camera both plug into it, so neither works without it. *Robu*
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS, 3 off	35,247	Runs the person-detection model fast enough to search at flying speed. *Robu*
Compute cooling	Official Raspberry Pi 5 active cooler, 3 off	1,557	The hottest moment is the setup on the ground, before the propellers are turning and cooling it. *Robu*
Storage	SanDisk High Endurance 128 GB microSD, 3 off	11,967	Records every frame and detection for a whole mission. Ordinary cards wear out at that rate. *Flipkart*
GNSS RTK receiver	Teravolt AeroNav-Pro RTK, 3 rover + 1 base, 4 off	1,00,000	Precise positioning: one per aircraft and one on the ground. The supplier has confirmed it works this way and quoted; this was an estimate before. *Teravolt*
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens, 3 off	20,397	The search sensor, chosen so a person in water is large enough in the picture to be found. *Robu*
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW, 3 off	12,717	Sends the live picture down, one channel per aircraft. *IndiaMART*
Command receiver	ExpressLRS RX24T, 2.4 GHz, 3 off	4,317	Carries the safety pilot's control and the aircraft's reporting on a single link, which saves a second radio. *Robu*
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm, 3 off	2,967	How the three aircraft tell each other what they have found. The most robust link on the aircraft, which is where survivor positions belong. *Robu*
Power module	Holybro PM07, 3 off	15,747	Reports battery voltage and current to the autopilot, which is what triggers a low-battery return. *Robu*
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in, 3 off	11,817	Powers the computer. Deliberately oversized: losing it in flight is the failure this line exists to prevent. *Amazon India*

Item	Model	INR, all 3	Why this part, and from whom
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC, 3 off	837	A separate supply for the servos, so a jammed servo cannot disturb the autopilot. *Robu*

Per aircraft, airframe and drive — phases 7–10, repeated at 17–20 and 25–28 INR 58,291 per aircraft, INR 1,89,539 in all

Item	Model	INR, all 3	Why this part, and from whom
Motors	Tarot TL96020, 5008, 340 KV, 12 off	56,436	Twelve. The first is measured on the stand before the other eleven are bought. *Robokits*
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair, 16 off	26,256	Sixteen for twelve positions. The item most often broken in flight testing, and the whole of our spares policy. *Robokits*
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A, 3 off	15,567	One board drives all four motors. Made in India, and previously a quotation. *Robu*
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm, 6 off	13,554	The four arms. Carbon, and comfortably strong — the joints, not the tubes, are what limit the airframe. *Robu*
Motor mounts	Tarot 25 mm motor mount, TL9602, 12 off	13,428	Joins motor to arm without machining the tube. This joint is the part that has to be right. *Robu*
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs, 3 off	9,036	Bolted rather than bonded, so an arm damaged in a heavy landing is swapped in the field instead of scrapping the frame. *Amazon India*
Landing gear	F450/F550 landing skid, 3 off	1,647	Gives clearance underneath for the kit magazine, and a stance wide enough for an untidy landing. *Robu*
Suspension springs	5 mm stainless double torsion spring, 32 off	64	Let the legs absorb a hard landing rather than pass it into the arms and the battery. *IndiaMART*
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	949	Filament for the magazine, mounts and covers. Holds its shape in the sun, which cheaper filament does not. *Robu*
Release servos	MG90S mini servo, 180 deg, 12 off	1,788	Open the four kit stations. Metal gears, because a plastic one can drop a kit without anyone knowing. *Robocraze*
Cells	Molicel INR21700-P45B, 6S3P, 54 off	27,486	The battery: enough for the mission, a second full search, and four minutes in hand. *Robu*
Cell holders	3x1 21700 holder, 21.75 mm bore, 54 off	648	Hold the cells in place and keep them apart, so one failing cell does not take its neighbours with it. *Robu*
Pack interconnect	Pure nickel strip, 0.15 x 27 mm	6,500	Joins the cells. Pure nickel, because the cheaper plated strip runs hot at the current this battery delivers. *IndiaMART*
Group interconnect	Tinned copper braided jumper	500	Joins the battery's groups, where the current is highest. *IndiaMART*
Balance leads	JST-XH 6S, 200 mm, 22 AWG, 3 off	372	How the charger checks each cell. With no battery board fitted, this is our only per-cell check — and it happens on every charge. *zBotic*
Pack fusing	ANL fuse holder + 150 A ANL fuses, 3 off	9,000	Protects against a short circuit without cutting power at full throttle. *Njour*
Pack retention	300 mm LiPo strap, reusable, 6 off	372	Holds the battery down. It is a fifth of the aircraft's weight, so a strap that lets go is not a small problem. *Robu*

Item	Model	INR, all 3	Why this part, and from whom
Main leads	10 AWG ultra-flexible silicone wire	149	The main power cable, sized for the highest current the motors ever draw rather than the average. *Robu*
Power connectors	Amass XT90S anti-spark, M/F pair, 12 off	2,160	Anti-spark, so connecting a battery this size does not burn the contacts. *Robokits*
Signal connectors	JST ZH 6-pin, 200 mm leads, 3 off	69	Keyed, so the wiring can only go back together one way after a repair. *Robu*
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters, 3 off	3,417	Let the antennas mount on the airframe instead of hanging off the radios. *Desertcart*
Insulation	PVC heat-shrink sleeve, 150 mm, 3 off	141	Sleeving over the battery joints — the one place on the aircraft where a rubbed wire is an immediate fire. *Robu*

Flight-test support — phases 11–12 INR 38,925 of parts

Item	Model	INR, all 3	Why this part, and from whom
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	16,519	Manual override for the safety pilot, independent of the autonomy. A rule requirement and a safety one. *zBotic*
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	7,349	Two batteries recharged between flights is what sets how many tests fit into a day. *Robu*
Pack health monitor	Holybro PM08-CAN, 14S, 200 A	9,058	A bench instrument for tracking how the batteries age over the programme. *Robu*
Cable management	Nylon zip ties, 350 mm, 100 pcs	135	Ties long enough to dress wiring around a one-metre airframe. *Robu*
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	159	Insulates the signal wiring during assembly. *Robu*
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	560	Holds radios and receivers where a screw would need a fixing the printed part cannot carry. *Amazon India*
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	145	For the battery and anything else that must come out between flights without tools. *Amazon India*
Consumables	Fasteners, tape, connectors, filament	5,000	Solder, abrasives, spare hardware, filament. A round figure, because itemising it would be false precision. *in-house*

Ground segment and statutory — phases 29–32 INR 19,436 of parts

Item	Model	INR, all 3	Why this part, and from whom
Video receivers	RC832S 5.8 GHz AV receiver, 3 off	13,197	Three, so the ground station can show all three aircraft at once, which the rules require. *Robu*
Receive antennas	Triple-feed 5.8 GHz patch, SMA, 3 off	2,517	The ground antennas. Most of the video range comes from these rather than from the transmitters. *Robu*
Video capture	USB 2.0 analog AV capture adapter, 3 off	1,197	Brings the three pictures into the ground station computer. *Robu*
Coordination base	EByte E22-900T22U, SX1262, USB	1,646	The ground end of the link the aircraft use to report what they find. *Hubtronics*
Base station mount	Photographic tripod and adapter	879	Holds the ground receiver still during a mission. Survey-grade accuracy is unnecessary; not moving is what matters. *Amazon India*

Why there is no battery management board. An earlier version funded one. Most of what that line covered is already bought under other names, and the part that remains — a board that cuts power when a cell is unhappy — is the wrong device on an aircraft, because it can cut power at full throttle, which is a crash rather than a protection. Instead the battery is fused against short circuits, the autopilot brings the aircraft home on low voltage, and every cell is checked on the charger before each flight. What we do not have is per-cell protection in the air; that is a deliberate choice, not an oversight.

What this arrangement costs us

Thirty approvals is a material administrative overhead, and buying one aircraft at a time places the same imported order three times — roughly eight to twelve weeks of schedule. Consolidating the per-aircraft phases into one order is available if schedule becomes binding.

The detection evaluation needs no procurement: about a week on a departmental GPU and 30 GB of storage, against a public dataset.

7. Future work

Near term. Measure what is currently calculated: detection accuracy, motor thrust, setup time. Evaluate near-infrared sensing — water absorbs strongly and skin does not, so a person appears bright against dark water. A camera without its infrared filter costs INR 3,000, and a public dataset allows evaluation before purchase. Only the training data is flood-specific; the same system suits earthquake, wildfire and avalanche search.

Longer term. A capable aircraft deciding for itself within a small power budget is a hard requirement in planetary aviation. *Ingenuity* flew 72 times on Mars from a 1.8 kg airframe in an atmosphere under one per cent of Earth's density; *Dragonfly* launches in 2028 for Titan. A command takes minutes to arrive, so every decision is made on board. What transfers is what is not flood-specific: onboard survey planning, perception run to an energy budget, and safe behaviour without a link. Microcontroller-class processors, often proposed for such missions, cannot resolve targets of this size, so a planetary configuration needs a different operating point.

8. Conclusion

Three aircraft, one operator, no network: survivors located in flooded terrain and reached with a relief kit. Eight minutes for ten hectares, returning coordinates a rescue coordinator can act on. The design is specified; what remains is to build it and measure it.

Exposure is never more than one phase. The first two total INR 9,014 and settle whether these motors lift this aircraft; if they do not, the programme stops there and the department keeps a thrust stand. Nothing above INR 26,000 is released until a measured thrust figure exists, and the two largest blocks come only after a complete aircraft has flown.

Floods recur every monsoon, and no capability of this kind exists at a scale two people can deploy. The whole system costs less than an hour of helicopter search time. We ask the department to release Phase 1.

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-1

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 1 of 30, Establish thrust and mass measurement capability

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 1 of 30: Establish thrust and mass measurement capability. This is the first step, and is gated only on this approval. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	1	1,140
Load cell	REES52 20 kg load cell + HX711 breakout	1	854
Threadlocker	Loctite 243, 50 ml	1	664
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	1	373
Load-cell amplifier	7Semi HX711 24-bit ADC	1	104
		Total	**3,135**

We kindly request approval and financial support of **INR 3,135** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-2

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 2 of 30, Confirm one motor lifts what the design needs

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 2 of 30: Confirm one motor lifts what the design needs. It falls due only once the previous step has produced its stated result: *instruments commissioned*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Motors	Tarot TL96020, 5008, 340 KV	1	4,703
		Total	**4,703**

We kindly request approval and financial support of **INR 4,703** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
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Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
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President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-3

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 3 of 30, Aircraft 1 – autopilot and control link

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 3 of 30: Aircraft 1 – autopilot and control link. It falls due only once the previous step has produced its stated result: *thrust verified*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Flight controller	Holybro Pixhawk 6C Mini	1	22,049
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	161
		Total	**22,210**

We kindly request approval and financial support of **INR 22,210** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
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Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

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THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
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Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-4

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 4 of 30, Aircraft 1 – onboard computer and AI accelerator

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 4 of 30: Aircraft 1 – onboard computer and AI accelerator. It falls due only once the previous step has produced its stated result: *autopilot bench-tested*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Companion computer	Raspberry Pi 5, 8 GB	1	19,999
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	11,749
Storage	SanDisk High Endurance 128 GB microSD	1	3,989
Compute cooling	Official Raspberry Pi 5 active cooler	1	519
		Total	**36,256**

We kindly request approval and financial support of **INR 36,256** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-5

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 5 of 30, Aircraft 1 – centimetre positioning (RTK rover)

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 5 of 30: Aircraft 1 – centimetre positioning (RTK rover). It falls due only once the previous step has produced its stated result: *compute stack operating*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
GNSS RTK receiver	Teravolt AeroNav-Pro RTK, 3 rover + 1 base	1	25,000
		Total	**25,000**

We kindly request approval and financial support of **INR 25,000** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-6

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 6 of 30, Aircraft 1 – camera and the three radios

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 6 of 30: Aircraft 1 – camera and the three radios. It falls due only once the previous step has produced its stated result: *position fix acquired*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	6,799
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW	1	4,239
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	1,439
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	989
		Total	**13,466**

We kindly request approval and financial support of **INR 13,466** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-7

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 7 of 30, Aircraft 1 – airframe fabrication

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 7 of 30: Aircraft 1 – airframe fabrication. It falls due only once the previous step has produced its stated result: *avionics integrated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	4,518
Motor mounts	Tarot 25 mm motor mount, TL9602	4	4,476
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3,012
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	1	949
Landing gear	F450/F550 landing skid	1	549
Suspension springs	5 mm stainless double torsion spring	10	20
		Total	**13,524**

We kindly request approval and financial support of **INR 13,524** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-8

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 8 of 30, Aircraft 1 – battery pack and power distribution

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 8 of 30: Aircraft 1 – battery pack and power distribution. It falls due only once the previous step has produced its stated result: *airframe fabricated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Cells	Molicel INR21700-P45B, 6S3P	18	9,162
Pack interconnect	Pure nickel strip, 0.15 x 27 mm	1	6,500
Power module	Holybro PM07	1	5,249
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3,939
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3,000
Group interconnect	Tinned copper braided jumper	1	500
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	279
Cell holders	3x1 21700 holder, 21.75 mm bore	18	216
Pack retention	300 mm LiPo strap, reusable	2	124
Balance leads	JST-XH 6S, 200 mm, 22 AWG	1	124
		Total	**29,093**

We kindly request approval and financial support of **INR 29,093** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

Ref.: RS/PH-9

Date: ____________

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 9 of 30, Aircraft 1 – speed controllers, release servos, wiring

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 9 of 30: Aircraft 1 – speed controllers, release servos, wiring. It falls due only once the previous step has produced its stated result: *pack bench-discharged.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A	1	5,189
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	1,139
Power connectors	Amass XT90S anti-spark, M/F pair	4	720
Release servos	MG90S mini servo, 180 deg	4	596
Main leads	10 AWG ultra-flexible silicone wire	1	149
Insulation	PVC heat-shrink sleeve, 150 mm	1	47
Signal connectors	JST ZH 6-pin, 200 mm leads	1	23
		Total	**7,863**

We kindly request approval and financial support of **INR 7,863** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-10

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 10 of 30, Aircraft 1 – propulsion completed

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 10 of 30: Aircraft 1 – propulsion completed. It falls due only once the previous step has produced its stated result: *drive train installed*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Motors	Tarot TL96020, 5008, 340 KV	3	14,109
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	4	6,564
		Total	**20,673**

We kindly request approval and financial support of **INR 20,673** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-11

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 11 of 30, Charging, pack instrumentation and manual override

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 11 of 30: Charging, pack instrumentation and manual override. It falls due only once the previous step has produced its stated result: *aircraft assembled and weighed.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	1	16,519
Pack health monitor	Holybro PM08-CAN, 14S, 200 A	1	9,058
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	1	7,349
		Total	**32,926**

We kindly request approval and financial support of **INR 32,926** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-12

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 12 of 30, Field consumables and assembly materials

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 12 of 30: Field consumables and assembly materials. It falls due only once the previous step has produced its stated result: *charging verified*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Consumables	Fasteners, tape, connectors, filament	1	5,000
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	1	560
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	1	159
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	1	145
Cable management	Nylon zip ties, 350 mm, 100 pcs	1	135
		Total	**5,999**

We kindly request approval and financial support of **INR 5,999** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-13

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 13 of 30, Aircraft 2 – autopilot and control link

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 13 of 30: Aircraft 2 – autopilot and control link. It falls due only once the previous step has produced its stated result: *aircraft 1 flew a full mission.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Flight controller	Holybro Pixhawk 6C Mini	1	22,049
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	161
		Total	**22,210**

We kindly request approval and financial support of **INR 22,210** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-14

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 14 of 30, Aircraft 2 – onboard computer and AI accelerator

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 14 of 30: Aircraft 2 – onboard computer and AI accelerator. It falls due only once the previous step has produced its stated result: *autopilot bench-tested*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Companion computer	Raspberry Pi 5, 8 GB	1	19,999
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	11,749
Storage	SanDisk High Endurance 128 GB microSD	1	3,989
Compute cooling	Official Raspberry Pi 5 active cooler	1	519
		Total	**36,256**

We kindly request approval and financial support of **INR 36,256** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR) | Signature | Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-15

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 15 of 30, Aircraft 2 – centimetre positioning (RTK rover)

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 15 of 30: Aircraft 2 – centimetre positioning (RTK rover). It falls due only once the previous step has produced its stated result: *compute stack operating*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
GNSS RTK receiver	Teravolt AeroNav-Pro RTK, 3 rover + 1 base	1	25,000
		Total	**25,000**

We kindly request approval and financial support of **INR 25,000** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-16

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 16 of 30, Aircraft 2 – camera and the three radios

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 16 of 30: Aircraft 2 – camera and the three radios. It falls due only once the previous step has produced its stated result: *position fix acquired*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	6,799
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW	1	4,239
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	1,439
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	989
		Total	**13,466**

We kindly request approval and financial support of **INR 13,466** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-17

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 17 of 30, Aircraft 2 – airframe fabrication

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 17 of 30: Aircraft 2 – airframe fabrication. It falls due only once the previous step has produced its stated result: *avionics integrated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	4,518
Motor mounts	Tarot 25 mm motor mount, TL9602	4	4,476
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3,012
Landing gear	F450/F550 landing skid	1	549
Suspension springs	5 mm stainless double torsion spring	10	20
		Total	**12,575**

We kindly request approval and financial support of **INR 12,575** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-18

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 18 of 30, Aircraft 2 – battery pack and power distribution

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 18 of 30: Aircraft 2 – battery pack and power distribution. It falls due only once the previous step has produced its stated result: *airframe fabricated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Cells	Molicel INR21700-P45B, 6S3P	18	9,162
Power module	Holybro PM07	1	5,249
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3,939
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3,000
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	279
Cell holders	3x1 21700 holder, 21.75 mm bore	18	216
Pack retention	300 mm LiPo strap, reusable	2	124
Balance leads	JST-XH 6S, 200 mm, 22 AWG	1	124
		Total	**22,093**

We kindly request approval and financial support of **INR 22,093** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-19

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 19 of 30, Aircraft 2 – speed controllers, release servos, wiring

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 19 of 30: Aircraft 2 – speed controllers, release servos, wiring. It falls due only once the previous step has produced its stated result: *pack bench-discharged*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A	1	5,189
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	1,139
Power connectors	Amass XT90S anti-spark, M/F pair	4	720
Release servos	MG90S mini servo, 180 deg	4	596
Insulation	PVC heat-shrink sleeve, 150 mm	1	47
Signal connectors	JST ZH 6-pin, 200 mm leads	1	23
		Total	**7,714**

We kindly request approval and financial support of **INR 7,714** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-20

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 20 of 30, Aircraft 2 – propulsion completed

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 20 of 30: Aircraft 2 – propulsion completed. It falls due only once the previous step has produced its stated result: *drive train installed*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Motors	Tarot TL96020, 5008, 340 KV	4	18,812
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	4	6,564
		Total	**25,376**

We kindly request approval and financial support of **INR 25,376** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR) | Signature | Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-21

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 21 of 30, Aircraft 3 – autopilot and control link

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 21 of 30: Aircraft 3 – autopilot and control link. It falls due only once the previous step has produced its stated result: *two aircraft, separation verified*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Flight controller	Holybro Pixhawk 6C Mini	1	22,049
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	161
		Total	**22,210**

We kindly request approval and financial support of **INR 22,210** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-22

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 22 of 30, Aircraft 3 – onboard computer and AI accelerator

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 22 of 30: Aircraft 3 – onboard computer and AI accelerator. It falls due only once the previous step has produced its stated result: *autopilot bench-tested*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Companion computer	Raspberry Pi 5, 8 GB	1	19,999
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	11,749
Storage	SanDisk High Endurance 128 GB microSD	1	3,989
Compute cooling	Official Raspberry Pi 5 active cooler	1	519
		Total	**36,256**

We kindly request approval and financial support of **INR 36,256** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-23

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 23 of 30, Aircraft 3 – centimetre positioning (RTK rover)

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 23 of 30: Aircraft 3 – centimetre positioning (RTK rover). It falls due only once the previous step has produced its stated result: *compute stack operating*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
GNSS RTK receiver	Teravolt AeroNav-Pro RTK, 3 rover + 1 base	1	25,000
		Total	**25,000**

We kindly request approval and financial support of **INR 25,000** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-24

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 24 of 30, Aircraft 3 – camera and the three radios

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 24 of 30: Aircraft 3 – camera and the three radios. It falls due only once the previous step has produced its stated result: *position fix acquired*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	6,799
Video transmitter	SunRobotics TS832, 5.8 GHz, 600 mW	1	4,239
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	1,439
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	989
		Total	**13,466**

We kindly request approval and financial support of **INR 13,466** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

Date

THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-25

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 25 of 30, Aircraft 3 – airframe fabrication

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 25 of 30: Aircraft 3 – airframe fabrication. It falls due only once the previous step has produced its stated result: *avionics integrated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	4,518
Motor mounts	Tarot 25 mm motor mount, TL9602	4	4,476
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3,012
Landing gear	F450/F550 landing skid	1	549
Suspension springs	5 mm stainless double torsion spring	12	24
		Total	**12,579**

We kindly request approval and financial support of **INR 12,579** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
Prabhjot Singh	Robotics and Artificial Intelligence Engg.	2024–2028	1024230021
Manan Kapoor	Computer Science & Engineering	2024–2028	1024030467
Manav Pansari	Electronics & Communication Engineering	2024–2028	1024060151
Ravi Kant Raja	Robotics and Artificial Intelligence Engg.	2024–2028	1024230032
Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
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President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

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THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-26

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 26 of 30, Aircraft 3 – battery pack and power distribution

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 26 of 30: Aircraft 3 – battery pack and power distribution. It falls due only once the previous step has produced its stated result: *airframe fabricated*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Cells	Molicel INR21700-P45B, 6S3P	18	9,162
Power module	Holybro PM07	1	5,249
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3,939
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3,000
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	279
Cell holders	3x1 21700 holder, 21.75 mm bore	18	216
Pack retention	300 mm LiPo strap, reusable	2	124
Balance leads	JST-XH 6S, 200 mm, 22 AWG	1	124
		Total	**22,093**

We kindly request approval and financial support of **INR 22,093** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
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Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
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President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

Amount sanctioned (INR)

Signature

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RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
Through: President, Thapar Amateur Astronomers Society (TAAS)
Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-27

Date: ____________________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 27 of 30, Aircraft 3 – speed controllers, release servos, wiring

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 27 of 30: Aircraft 3 – speed controllers, release servos, wiring. It falls due only once the previous step has produced its stated result: *pack bench-discharged*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Speed controllers	Darkmatter VISHNU 50 MK-III, 4-in-1, 50 A	1	5,189
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	1,139
Power connectors	Amass XT90S anti-spark, M/F pair	4	720
Release servos	MG90S mini servo, 180 deg	4	596
Insulation	PVC heat-shrink sleeve, 150 mm	1	47
Signal connectors	JST ZH 6-pin, 200 mm leads	1	23
		Total	**7,714**

We kindly request approval and financial support of **INR 7,714** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
Samuel Masih	Computer Science & Engineering	2024–2028	1024170044
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Neil Kishore	Mechanical Engineering	2024–2028	1024080017

Approvals & Signatures

Dr. Surbhi Sharma
Mentor

Dr. Mamata Gulati
Mentor

President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

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Date

RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
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Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-28

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 28 of 30, Aircraft 3 – propulsion completed

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 28 of 30: Aircraft 3 – propulsion completed. It falls due only once the previous step has produced its stated result: *drive train installed*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
Motors	Tarot TL96020, 5008, 340 KV	4	18,812
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	8	13,128
		Total	**31,940**

We kindly request approval and financial support of **INR 31,940** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

Name	Department	Batch	Roll No.
Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
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Approvals & Signatures

Dr. Surbhi Sharma
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Dr. Mamata Gulati
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President, TAAS
Thapar Amateur Astronomers Society

Swastik Kumar
Department of Electronics & Communication Engineering, Roll No. 1024060150
Submitted on behalf of the team

Date

For office use — Office of the Dean, Student Affairs

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RescueSwarm — An Autonomous Multi-UAV System for Post-Disaster Search, Localisation and Payload Delivery

To
The Dean, Student Affairs (DoSA)
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Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-29

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 29 of 30, RTK base receiver, the source of the corrections

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 29 of 30: RTK base receiver, the source of the corrections. It falls due only once the previous step has produced its stated result: *third aircraft flying*. We request funding for the following components:

Component	Model	Qty	Cost (INR)
GNSS RTK receiver	Teravolt AeroNav-Pro RTK, 3 rover + 1 base	1	25,000
		Total	**25,000**

We kindly request approval and financial support of **INR 25,000** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

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Approvals & Signatures

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Swastik Kumar
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Submitted on behalf of the team

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THAPAR AMATEUR ASTRONOMERS SOCIETY

THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY (Deemed to be University)

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Thapar Institute of Engineering & Technology, Patiala

Ref.: RS/PH-30

Date: ____________

Subject: Fund Request under TAAS Society for the RescueSwarm Project — Phase 30 of 30, Ground station: three video feeds, data link, base mount

Respected Ma'am,

We, a team of students of Thapar Institute of Engineering & Technology, are working under the guidance of Dr. Surbhi Sharma and Dr. Mamata Gulati on **RescueSwarm**, an autonomous multi-UAV system for post-disaster search, survivor localisation and payload delivery. The software and simulation framework is complete, and the programme is now in hardware integration and testing, funded in 30 steps so that each is approved only once the previous has produced a stated result.

This request is Phase 30 of 30: Ground station: three video feeds, data link, base mount. It falls due only once the previous step has produced its stated result: *base receiver acquired.* We request funding for the following components:

Component	Model	Qty	Cost (INR)
Video receivers	RC832S 5.8 GHz AV receiver	3	13,197
Receive antennas	Triple-feed 5.8 GHz patch, SMA	3	2,517
Coordination base	EByte E22-900T22U, SX1262, USB	1	1,646
Video capture	USB 2.0 analog AV capture adapter	3	1,197
Base station mount	Photographic tripod and adapter	1	879
		Total	**19,436**

We kindly request approval and financial support of **INR 19,436** under TAAS Society for the procurement of the components above. This is the parts cost of this phase; the programme schedule additionally carries tax where it is still owed, and a 15 % contingency.

Team Members *Mentors:* Dr. Surbhi Sharma and Dr. Mamata Gulati

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Swastik Kumar	Electronics & Communication Engineering	2024–2028	1024060150
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Approvals & Signatures

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Swastik Kumar
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